Overnight SPY hedge assessment

Executive report · 2026-10-30

Verdict

Buying 500 SPY shares at the expiry close limits modeled overnight exposure to plus or minus 500 shares, but did not produce consistent profits after modeled costs. Mean net profit per package was -$190.13 in 2022, -$211.54 in 2024, and +$43.21 in 2025. Each descriptive 95% interval includes zero. In 2024, it improved the mean by $296.56 versus no hedge, but the mean remained negative.

The rule based on that morning's stock price averaged -$136.85 in 2022, -$90.03 in 2024, and +$653.39 in 2025. Its 2025 descriptive interval was +$125.64 to +$1,274.85. The median was -$10.08. It left modeled stock exposure on 48 of 127 expiries, and the overnight stock-price contribution averaged +$777.20. The positive result still involved stock risk.

The diagnostic that removes all modeled overnight shares lost money after the main cost assumptions in every period. It requires future assignment information and cannot be used as a trading rule or a guaranteed profit bound.

These are historical comparisons on previously examined periods. Live execution remains untested.

The prespecified 500-share hedge leaves residual exposure of plus or minus 500 shares. This report makes no operational profit claim.

One package is ten short SPY call contracts and one long SPXW call contract, each with a 100 multiplier. The table shows the stock hedges applied to that package.

Mean net USD per package by period

Period	n	No hedge	500-share hedge	1,000-share hedge	Morning-price rule
2019	9	-$827.24	-$792.27	-$757.30	-$610.75
2022	63	$96.47	-$190.13	-$476.73	-$136.85
2024	195	-$508.10	-$211.54	$85.02	-$90.03
2025 (Jan-Aug)	127	$191.97	$43.21	-$105.54	$653.39

2025 covers January through August. Full-cross, base costs, gross prefunding, 1 bp slippage. The other 12 period/scenario rows are in Markdown.

Primary uncertainty and sensitivity

The 500-share hedge was prespecified. Its absolute and paired 95% intervals use full-cross, gross prefunding, and 1 bp. No-hedge absolute intervals are shown beside them. The 2019 pilot has means only.

Period	No-hedge mean	No-hedge abs CI	500-share mean	500-share abs CI	Paired vs no hedge	Paired CI
2022	$96.47	-$166.32 to $347.98	-$190.13	-$591.85 to $255.63	-$286.60	-$532.06 to $17.88
2024	-$508.10	-$795.19 to -$237.03	-$211.54	-$447.40 to $10.25	$296.56	$105.71 to $479.20
2025 (Jan-Aug)	$191.97	-$406.29 to $854.57	$43.21	-$279.14 to $356.71	-$148.75	-$768.07 to $374.28

Base-cost intervals are descriptive. Stress intervals and the full uncertainty matrix are in Markdown.

Sensitivity uses the prespecified 500-share hedge. No strategy winner is selected.

Period	Base 0bp	Base 1bp	Base 5bp	Stress 0bp	Stress 1bp	Stress 5bp
2022	-$151.03	-$190.13	-$346.51	-$245.21	-$284.31	-$440.70
2024	-$157.42	-$211.54	-$428.03	-$279.61	-$333.73	-$550.22
2025 (Jan-Aug)	$102.56	$43.21	-$194.18	-$22.16	-$81.51	-$318.89

Funding assumption contrast at 1 bp, full-cross

Period	n	Gross base	Gross stress	Cash-net base	Cash-net stress
2022	63	-$190.13	-$284.31	-$168.63	-$241.31
2024	195	-$211.54	-$333.73	-$175.92	-$262.49
2025 (Jan-Aug)	127	$43.21	-$81.51	$76.82	-$14.30

Gross prefunding charges funding on the full stock purchase. Assignment-cash net credits modeled assignment cash against that need. This is an optimistic accounting contrast, not proof of broker settlement or timely access to funds.

Capital, residual exposure, and costs

Buy at the close, then sell a residual long or buy back a residual short at the next open. End flat. The morning-price rule sizes from the expiry-session SPY open and frozen SPY strike, known before the closing decision. It does not use assignment or the expiry close. Availability of a later order is unverified.

2024 risk summary, n=195. Other periods are in Markdown.

Hedge case	Gross buy USD	Max notional USD	Cash drawdown USD	Residuals	Long	Short	Residual SD
No hedge	$0.00	$607,690.00	$103,924.65	114	0	114	494.1
500-share hedge	$303,905.00	$303,905.00	$47,887.24	195	81	114	494.1
1,000-share hedge	$607,810.00	$607,810.00	$29,122.70	81	81	0	494.1
Morning-price rule	$607,810.00	$607,810.00	$30,912.20	50	27	23	507.3

Gross stock-purchase notional and maximum stock notional do not measure total required capital or margin. A zero no-hedge purchase does not mean zero total capital. Peak-to-trough drawdown is cash drawdown. Residuals may be long or short; partial assignment is unobserved.

Oracle diagnostic

Uses future assignment information. It is a separate diagnostic, not a tradable rule, not ranked with candidate strategies, and not a guaranteed-profit upper bound.

Period	n	Mean USD	95% CI
2019	9	-$63.75	Means only
2022	63	-$85.27	-$134.93 to -$42.02
2024	195	-$71.93	-$101.81 to -$43.30
2025 (Jan-Aug)	127	-$90.96	-$214.95 to -$4.56

Cost assumptions

Options: 11 contracts at $0.65 base and $1.50 stress. Stock: $0.005/share base, $0.010/share stress. Borrow: 3.0%/10.0%. Funding: 5.0%/10.0%. Slippage: 0, 1, 5 bp. Dividends: Unconditional cash dividend equals residual shares multiplied by the dividend per share.

Historical caveats

The 500-share hedge is partial. The 1,000-share hedge can create uncovered long exposure. Daily OHLC does not prove fills, capacity, or margin. No margin estimate is supplied.

Prior study and reproducibility

The reviewed stock-reference study found no dependable executable edge. It compared stock-reference inputs while holding the option sample fixed. The overnight hedge study answers a separate after-cost question.

The original matched-moneyness gap and expiry carry answer different questions. The stock-source comparison did not create an edge.

The primary case is full-cross, base costs, gross prefunding, and 1 bp slippage. The option and stock reference use a one-calendar-day expiration design with modeled buys at the daily close. Residual stock is closed at the next session open, leaving the modeled hedge flat.

These are descriptive historical comparisons on periods already examined, not a predictive holdout. Option quotes from 15:59 ET are used for 2024 and 2025 only and are paired with later close references. Clock semantics for 2019 and 2022 remain unresolved. The assignment threshold is a proxy; partial exercise is unobserved. Daily OHLC prices do not prove order availability, fills, capacity, or margin. No Sharpe, ROI, or operational profit claim is made.

DTE 1 quotes do not provide same-day delta or closeout.

The 36 no-hedge, zero-slippage summary controls reproduce the earlier study. This report's main table uses 1 bp stock slippage; the 2022 no-hedge mean of $96.47 versus the earlier $114.07 reflects that added assumption, not a source-data change.

Run record	Value
Evaluation	EVALUATED
Evaluated SHA prefix	db5d980c7abc
Clean SHA state	True
Run manifest SHA prefix	237c58999379
Protocol SHA prefix	30eae8115233
Normalized selection SHA prefix	8156d9030ca0
Input manifest SHA prefix	576d58194e46
Validation result / index prefixes	640193de6b4c / 42ba3e479450
Independent validation	PASS / 2,361,472 comparisons / 0 mismatches
Full repository tests	444 passed (206 warnings)
Focused overnight hedge tests	26 passed
Run start UTC	2026-10-29T17:18:26.473822+00:00
Run end UTC	2026-10-29T17:18:32.167674+00:00

Full hashes, command, runtime, and bound source list are in the Markdown companion and render manifest.

OIC assignment and exercise

NYSE opening and closing auctions